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Coimbatore, Tamilnadu

AJAY KARTHEEK

ELECTRONICS AND COMMUNICATION ENGINEERING

Professional Goals

Passionate about developing solutions that solve problems and make life better by leveraging AI, Gen AI, Electronics and IoT in conjunction with Cloud hyperscalers. Aiming to utilize my skills, continuously learn and innovate to further the mission of the organization I will be associated with.

Education

Bachelor Of Engineering, Electronics & Communication
Sri Shakthi Institute of Engineering And Technology, Coimbatore | 2021 - Present

Higher Secondary Education, Computer Science
RJ Matric Hr. Sec. School, Coimbatore | 2019 - 2021

Secondary Education
Amrita Vidyalayam CBSE, Coimbatore | 2019

Skills

PYTHON

GEN AI

Langchain

AWS

Programming, Frameworks And Tools

- Programming Skills: Python.
- Gen AI Models: Azure Open AI, Llama, Gemini Pro, Open AI's GPT 3, GPT 4 and Dall-E
- LLM Orchestrators: LangChain, Prefect
- Framework and Tools: Guardails.ai, Langraph, Google Collab, Jupyter NB, GitHub Copilot, CopilotKit.AI
- Hyperscalers: AWS Cloud Platform

Certificates And Training

- 450+ Leetcode Problems Solved and achieved rank of 10000 globally.
- IIT Kharagpur AI4ICPS (AI for Real world applications) Certificate Programme (On-going)
- GPT and Langchain - Developer By Stepan Grider
- Python Bootcamp Course by Jose Portila on Udemy
- Introduction to Gen AI, Google Skillboost
- AWS Cloud Practitioner Course by Stephan Maarek on Udemy
- IoT with Robotics Workshop by IIT Madras
- NVIDIA-Certified Associate Generative AI LLMs Training (On-going)
- Python Data Structures and Algorithm On PrepInsta

Key Projects

GEN AI - Chat With MySQL Database Using GPT-4 And Mistral AI

Developed a dynamic chat interface integrated with MySQL database using GPT-4, Mistral AI, LangChain, and Streamlit within a Python environment. Orchestrated the project from conceptualization to implementation, orchestrating seamless communication between the chatbot and database. Leveraged advanced AI technologies to enhance conversational capabilities and streamline data retrieval processes. This project showcases adept proficiency in AI integration, database management, and user interface development, culminating in an interactive and efficient solution for data-driven conversations.

Technology Platforms: Langchain, Open AI's GPT 3.5 turbo, Local MySQL DB, Streamlit GUI
LinkedIn URL - Talk To Database

GEN AI - Powered Blogging Platform

Developed AIBlogging, an interesting AI-powered blogging platform designed to revolutionize content creation and consumption. Utilizing cutting-edge technologies such as Next.js, LangChain, OpenAI, Tavily, and CopilotKit, the platform offers a sleek and intuitive interface for a seamless user experience across devices. AIBlogging stands out with its integration of OpenAI's Language Model (LLM), enabling users to effortlessly generate high-quality content.
Additionally, the incorporation of LangChain & Tavily provides an intelligent web-searching AI agent that curates relevant content, drastically reducing research time. CopilotKit further enhances the user experience by simplifying AI integration, making powerful AI tools accessible to both novice & experienced bloggers.

Technology Platforms: Langchain, Open AI's GPT 4 model, Tavily AI, Copilotkit.AI, Next.js
LinkedIn URL - AI Powered Blogging Platform

GEN AI - Resume Analyzer & Classifier

An innovative Resume Analyzer project leveraging Gen AI, which enables recruiters to short list profiles based on the job requirements as well as candidates to analyze their resumes, get recommendations and improve to get shortlisted. This solution allows to dynamically choose the models between OpenAI and Google Gemini AI as well as the language choices. The architecture is based on Langchain for integration, Streamlit for GUI, Gemini and OpenAI LLMs.
The application harnesses the power of natural language processing and machine learning to evaluate the content, structure, and overall impact of resumes, offering personalized feedback and suggestions for enhancement. The user interface is designed with the Streamlit framework, ensuring an intuitive, configurable and user-friendly experience for end-users.

Technology Platforms: Open AI's GPT-3.5 turbo, GPT-4 turbo, Google Gemini Pro, Cohere, FAISS, Streamlit GUI, Langchain
LinkedIn URL - Resume Analyser Post

GEN AI & RAG Powered Document Summarization, Comparison & QnA

Design and Development of a Documents QnA bot using Gen AI. Architecture with Langchain OpenAI embedding GPT-3 technology with RAG for accurate understanding and responses. Intuitive chat interface for effortless interaction. Lightning-fast search capabilities to retrieve information in seconds. Eg: Resume data extraction, Bill comparison, Legal docs QnA etc.
Chain of thoughts have been enabled with conversational buffer which enables natural language conversations with the context of previous answers.

Technology Platforms: Langchain, FAISS, Open AI Model, Streamlit GUI, PyPDF Loader
Deployed at Render - Website URL : https://pdf-chatter-ajay.onrender.com

GEN AI & RAG Powered Website Scrapping & QnA

Developed an innovative website scraping and Q&A bot powered by RAG and GPT models, LangChain technology, OpenAI embeddings, ChromeDB vector database, and Streamlit framework. Design and implementation of the system architecture, ensuring seamless integration of cutting-edge technologies. Achieved high accuracy and efficiency in answering user queries by leveraging advanced natural language processing techniques.
The application allows users to provide the URLs from the which the answers have to retrieved and the solution provides response for questions based on the information scraped form the provider URL in the real time.

Technology Platforms: Langchain, FAISS, Beautiful Soup, ChromeDB, Streamlit GUI
Deployed at Render - Website URL : https://webpage-bot-query.onrender.com

IoT Controller for Gate Automation Using Node MCU and Integration with AWS

Developed a smart garage door controller system integrating AWS cloud services, enabling remote access and monitoring via a smartphone application. Spearheaded the entire project lifecycle from conception to deployment, implementing AWS IoT Core for secure communication and real-time status updates. Prioritized user experience, ensuring intuitive controls and seamless functionality while maintaining robust security measures including encrypted communication and user authentication. The system's scalability and reliability were ensured through AWS's scalable infrastructure, offering homeowners enhanced convenience and security in managing their garage doors remotely.

Internship

Kovai Digitech
Coimbatore | 1 Month

Participated in an internship with a focus on CCTV cameras and home automation, gaining practical insights into the setup and functionality of surveillance systems. Contributed to projects by exploring and implementing solutions for smart home automation and video monitoring

Enthu Technology
Coimbatore | 2 Weeks

Participated in a dynamic IoT internship, applying theoretical knowledge to hands-on projects. Explored innovative solutions for real-world applications.

Tech UC
Coimbatore | 2 Weeks

Interned with a focus on Arduino controllers, putting practical skills to use for real-world projects. Explored and implemented creative solutions within the Arduino technology space

Personal Details

Date of Birth : 18-06-2003

Father's Name : Srinivas S P

Mother's Name : Vijayalakshmi M S

Permanent Address : 16/2 Ganeshar Street, Subbenaiken Pudur, Coimbatore - 641049

Gender : Male

Hometown : Coimbatore

Mobility : Willing to relocate anywhere in India or abroad

Declaration

I hereby declare that all the information contained in this resume is by facts or truths to my knowledge. I take full responsibility for the correctness of the said information.

- Ajay Kartheek